

AForce OS — Patent Figures

AForce OS — Patent Figures (FIG. 1 - FIG. 10)

Companion to: aforce-os-engineering-brief.md **Audience:** Outside counsel and USPTO filing **Date:** April 29, 2026

This document reproduces the ten patent figures referenced throughout the engineering brief, in filing order, one figure per page, with a standardized caption block.

Each figure is also available as:

- **Vector SVG** — fig{1..10}~*.svg (editable, zero-loss)
- **High-resolution PNG** — png/fig{1..10}~*.png (2000 px wide, ~300 DPI at filing size)
- **Embedded inline** — at the matching section of the engineering brief

FIG. 1 — System Architecture

Maps to: Brief §1 (Mobile Application)

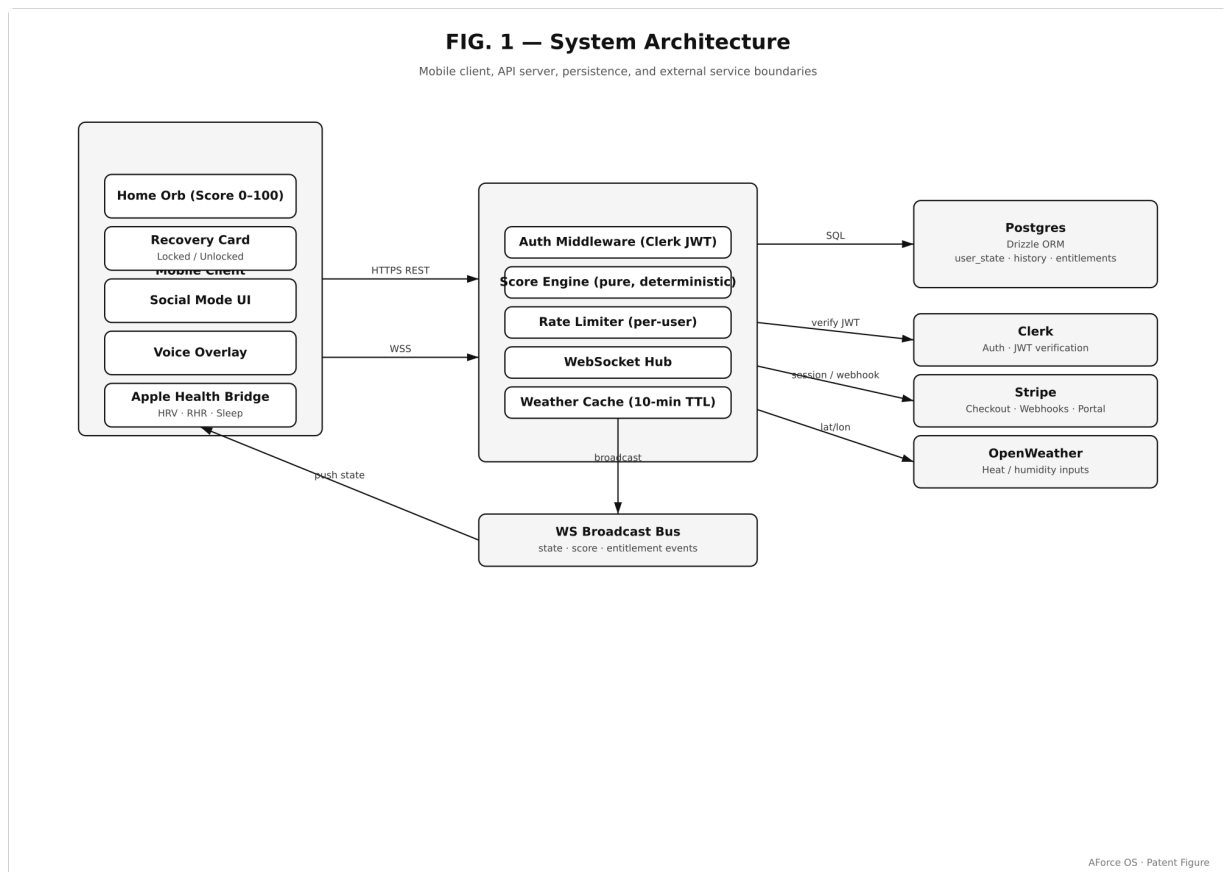


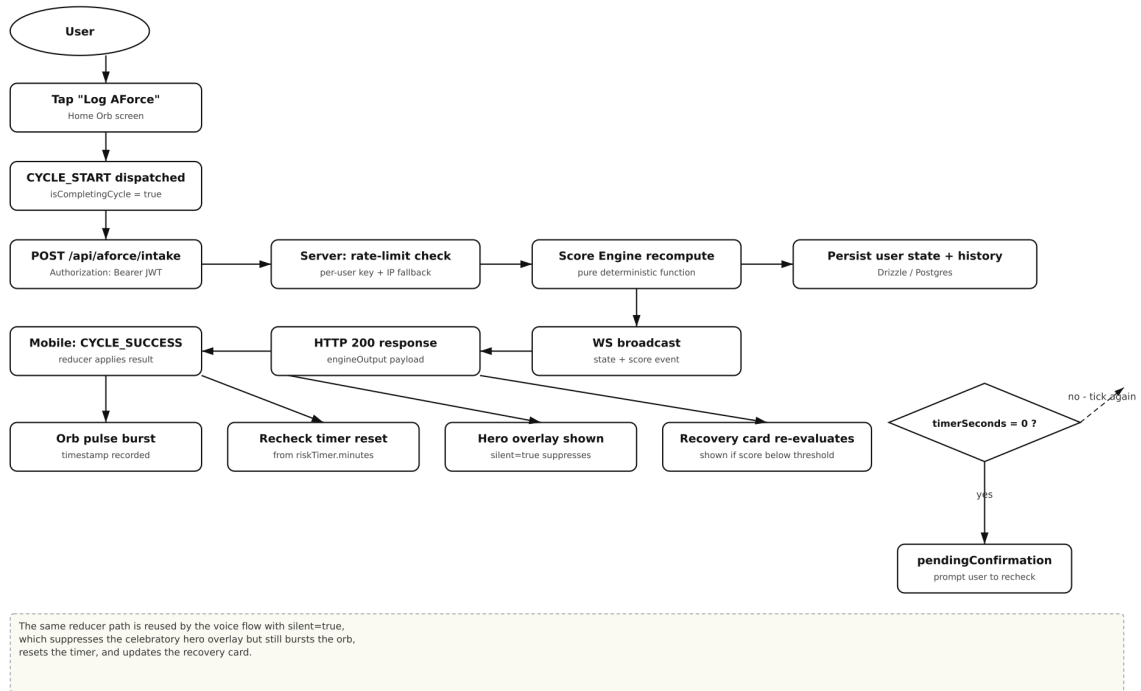
FIG. 1 — System Architecture. Top-level diagram of the AForce OS mobile client (Expo / React Native), the api-server (Express + Drizzle), the Postgres data store, and the WebSocket realtime channel that synchronizes state across devices.

FIG. 2 — Hero Intake Flow

Maps to: Brief §1.1 (home screen)

FIG. 2 - Hero Intake and Score Recompute Cycle

User logs an AForce intake; orb pulse, recheck timer, and recovery banner update in lockstep



AForce OS - Patent Figure

FIG. 2 — Hero Intake Flow. The Hydration Control Center loop: animated status orb (band-coloured) → AI command (WHAT + WHEN + OUTCOME) → user logs intake (one tap or voice) → score recompute via the pure scoring engine → orb re-renders, prediction strip updates, recheck timer resets to the new cadence.

FIG. 3 — Paywall Flow

Maps to: Brief §1.4 (entitlement)

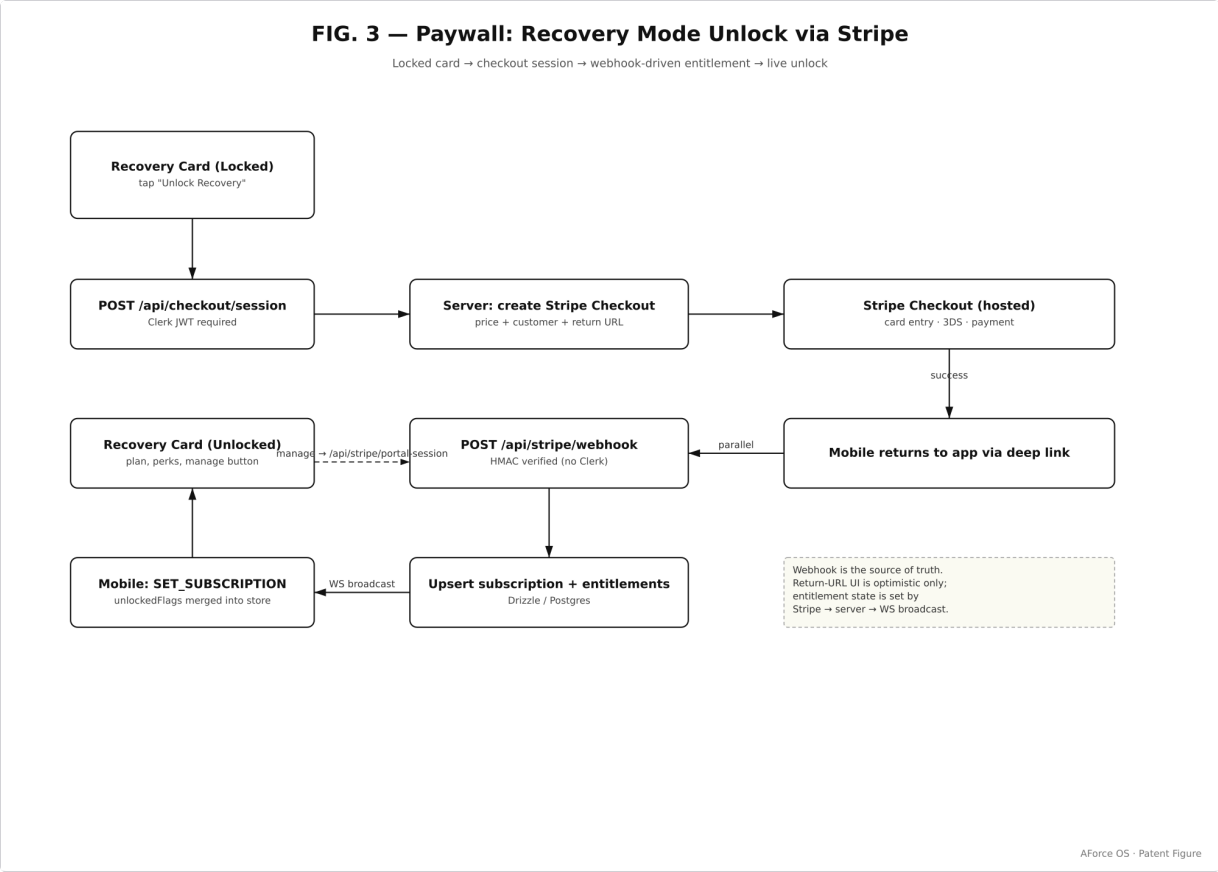


FIG. 3 — Paywall Flow. Premium gating for Recovery Card, Sweat Calculator, Voice, Heat Guard, and Autoscan. The `useEntitlement` hook checks the persisted subscription flag and a server-confirmed entitlement; the locked variant of each component renders an in-context upsell that preserves the user's session state.

FIG. 4 — Social Mode Flow

Maps to: Brief §3.3 (Social Mode override)

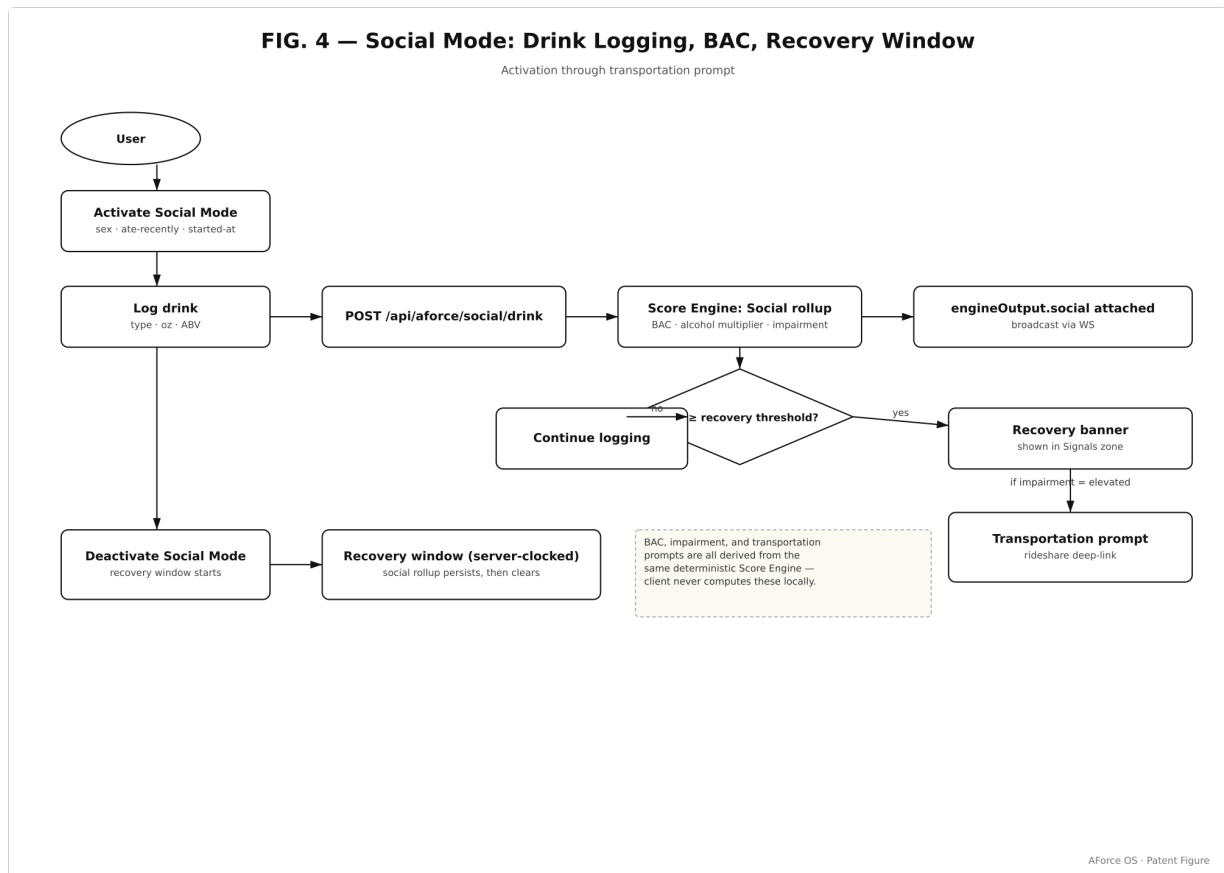


FIG. 4 — Social Mode Flow. Drink-logging panel → Widmark BAC estimate (parameterized by sex, weight, drinks, ABV, time elapsed, food intake) → impairment-level mapping (LOW / ELEVATED / MODERATE / HIGH / CRITICAL) → safety-class command override that preempts the standard performance ladder. Disclaimer copy is locked.

FIG. 5 — Voice Command Flow

Maps to: Brief §1.4 (voice)

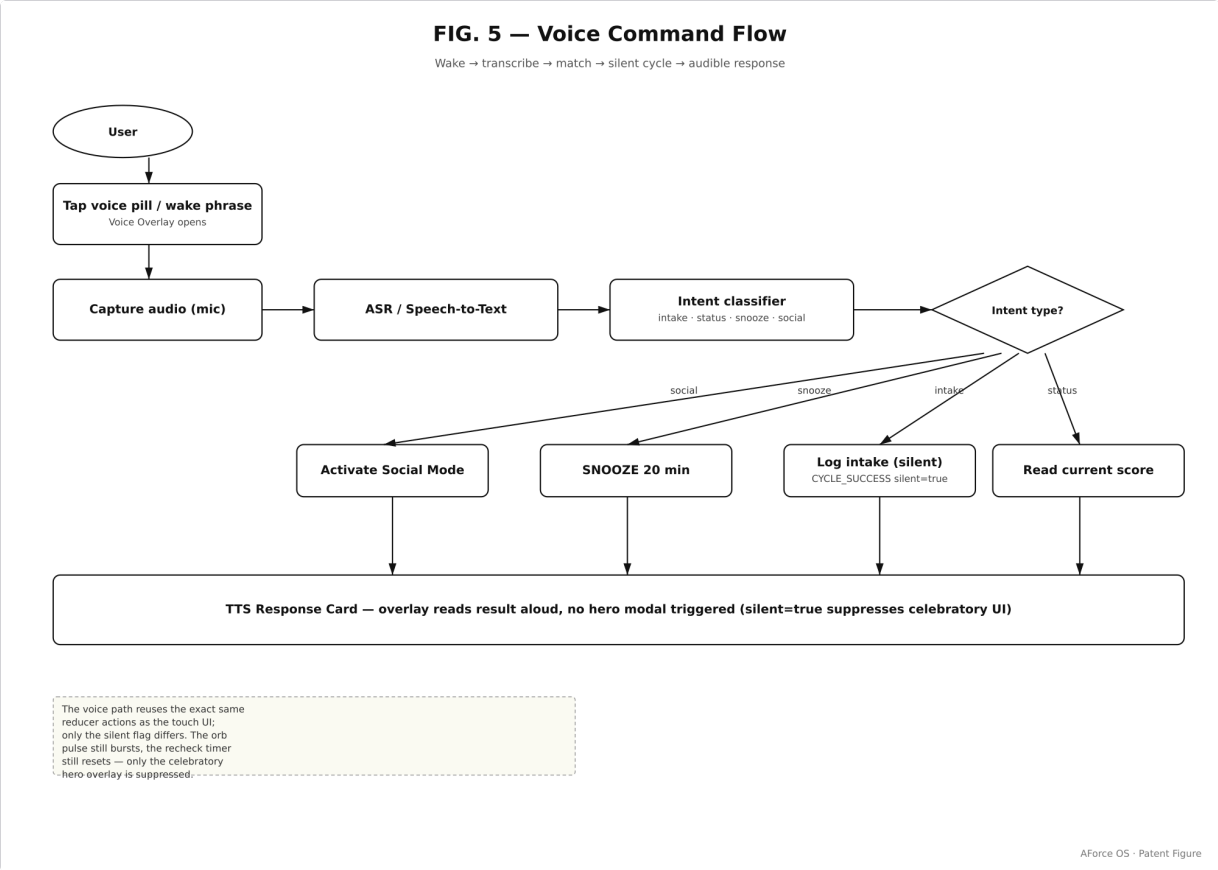


FIG. 5 — Voice Command Flow. Audio transcript → intent classification (LOG_INTAKE, GET_STATUS, GET_PREDICTION, START_SWEAT, OPEN_SCAN, ...) → action dispatch into the same reducer that handles tap actions → spoken response rendered by the brand template engine. The voice surface is a pure UI on top of the same state machine — there are no voice-only branches.

FIG. 6 — Wireframes

Maps to: Brief §1.5 (wireframes)

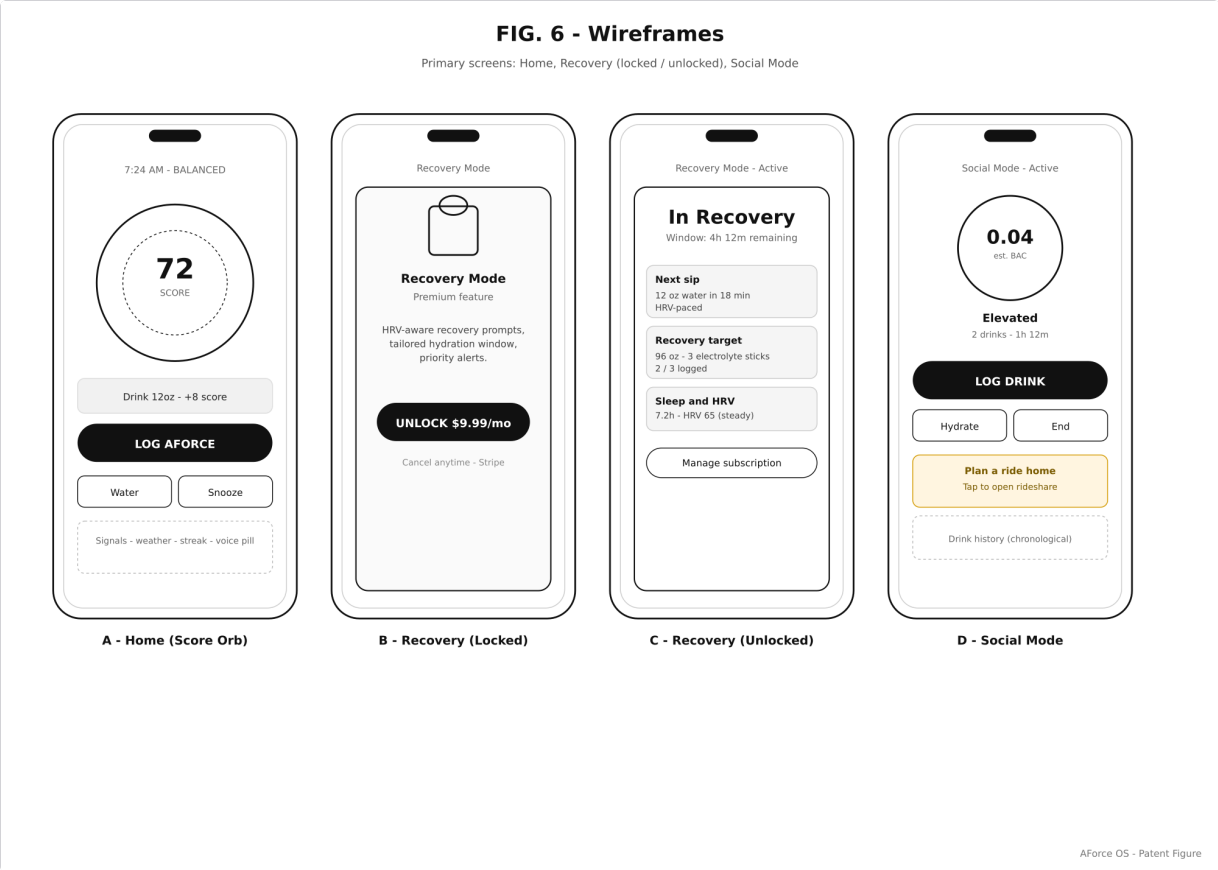


FIG. 6 — Wireframes. The five primary tabs of AForce OS — Hydration Control Center (home), Performance Signals (check), Recovery Protocol, Store, and Profile — plus the Sweat Calculator and Autoscan modal entry points. All wireframes are also available as live, interactive iframes on the engineering canvas.

FIG. 7 — Mobile App Architecture

Maps to: Brief §1.2 (state management)

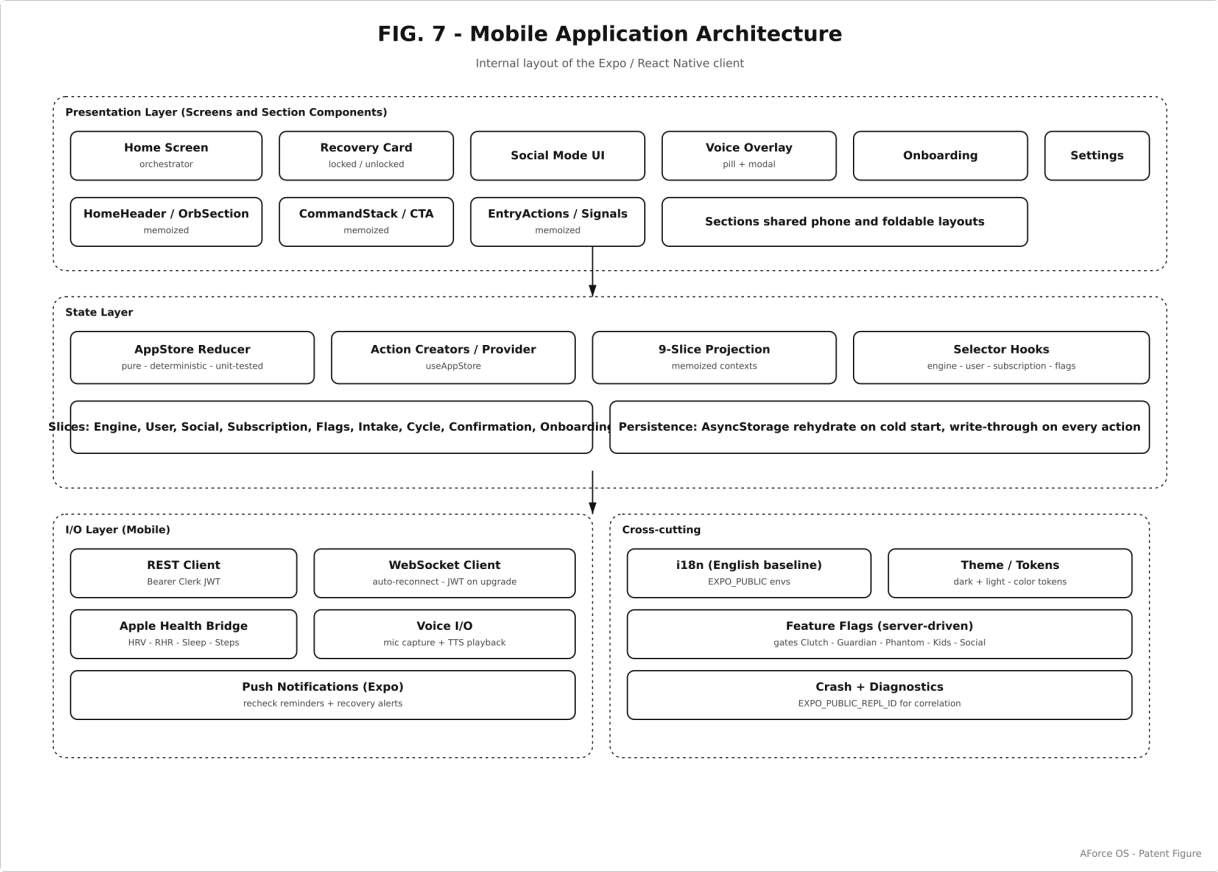


FIG. 7 — Mobile App Architecture. Sliced React Context + pure reducer + memoized slice projections so unrelated UI sections do not re-render on unrelated state changes. Expo-router file-based navigation. Clerk auth bridge mediates between Clerk session and the internal store. Apple HealthKit signals enter the engine through the `appleHealth` field of `UserState`.

FIG. 8 — Scoring Engine

Maps to: Brief \$2 (Scoring Engine)

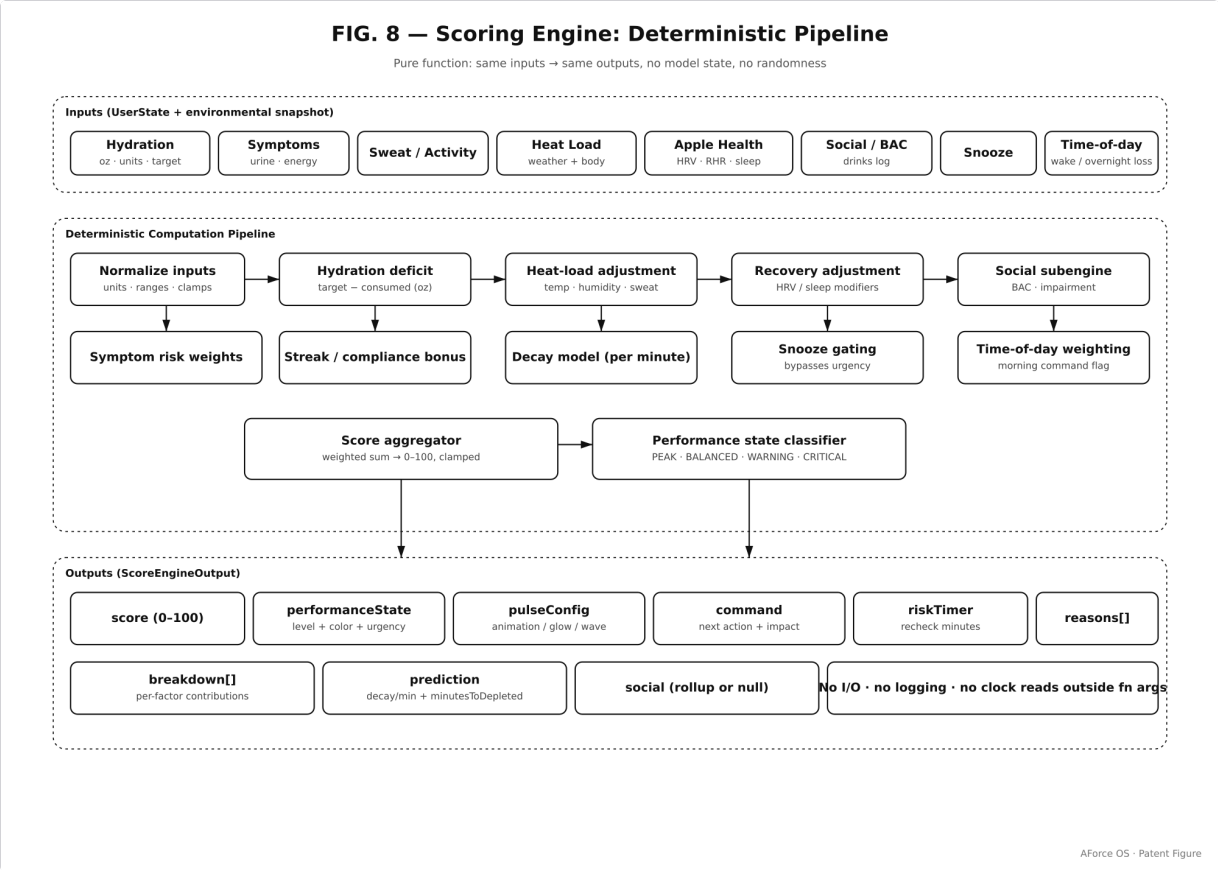


FIG. 8 — Scoring Engine. Pure-function pipeline from UserState snapshot through (a) per-event absorption, (b) bounded contribution sum, (c) continuous decay model, to (d) score / band / breakdown / prediction / command. Server- and client-side execution produce identical output for identical input ("score identity invariant").

FIG. 9 — Deterministic AI Layer

Maps to: Brief §3 (Deterministic AI Layer)

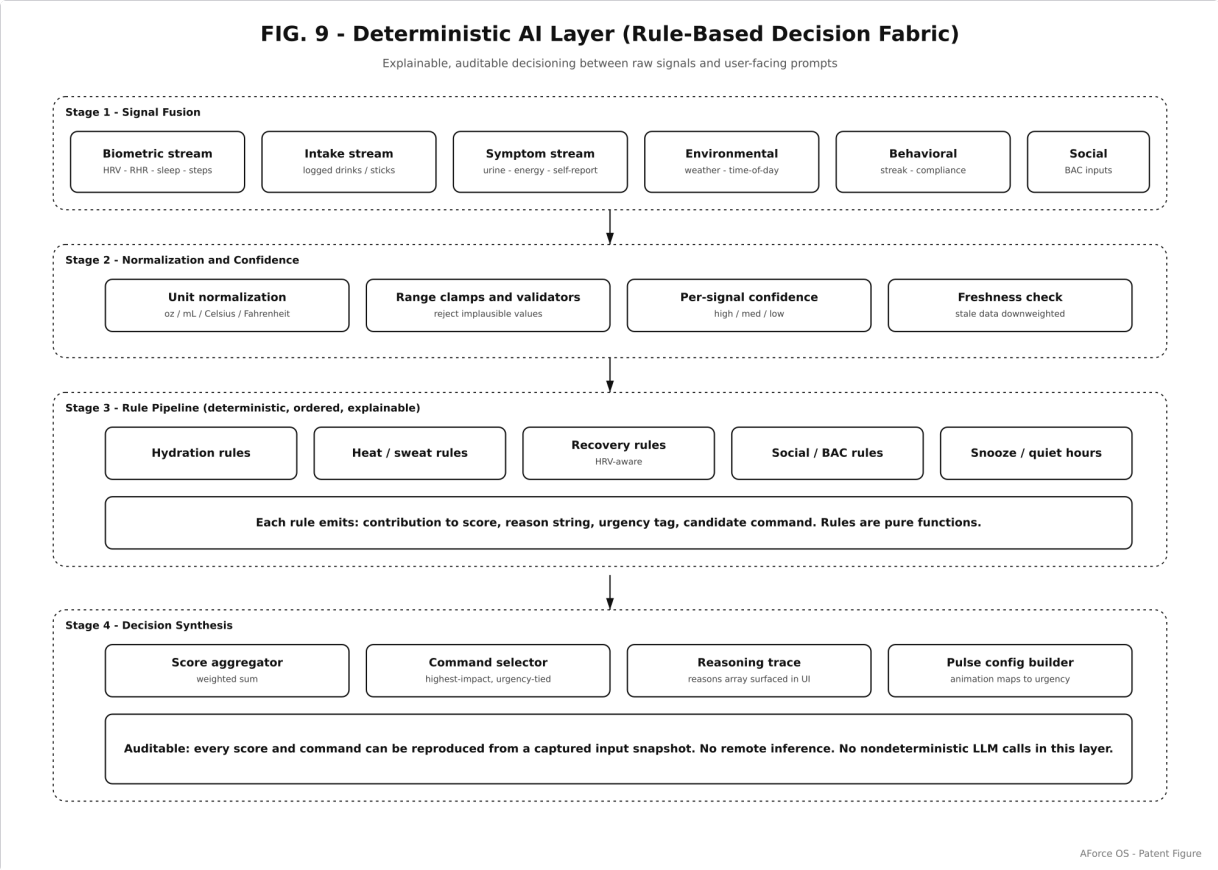


FIG. 9 — Deterministic AI Layer. Ordered rule pipeline: mode classification → penalty application → bonus application → decay projection → command generation → recheck cadence. No remote LLM inference in the safety- and performance-critical path; the same snapshot always produces the same command. Safety-class commands (Social Mode, Guardian Risk Engine) preempt the standard band ladder.

FIG. 10 — Autopilot / Autoscan

Maps to: Brief §4 (Autopilot) and §5 (Autoscan)

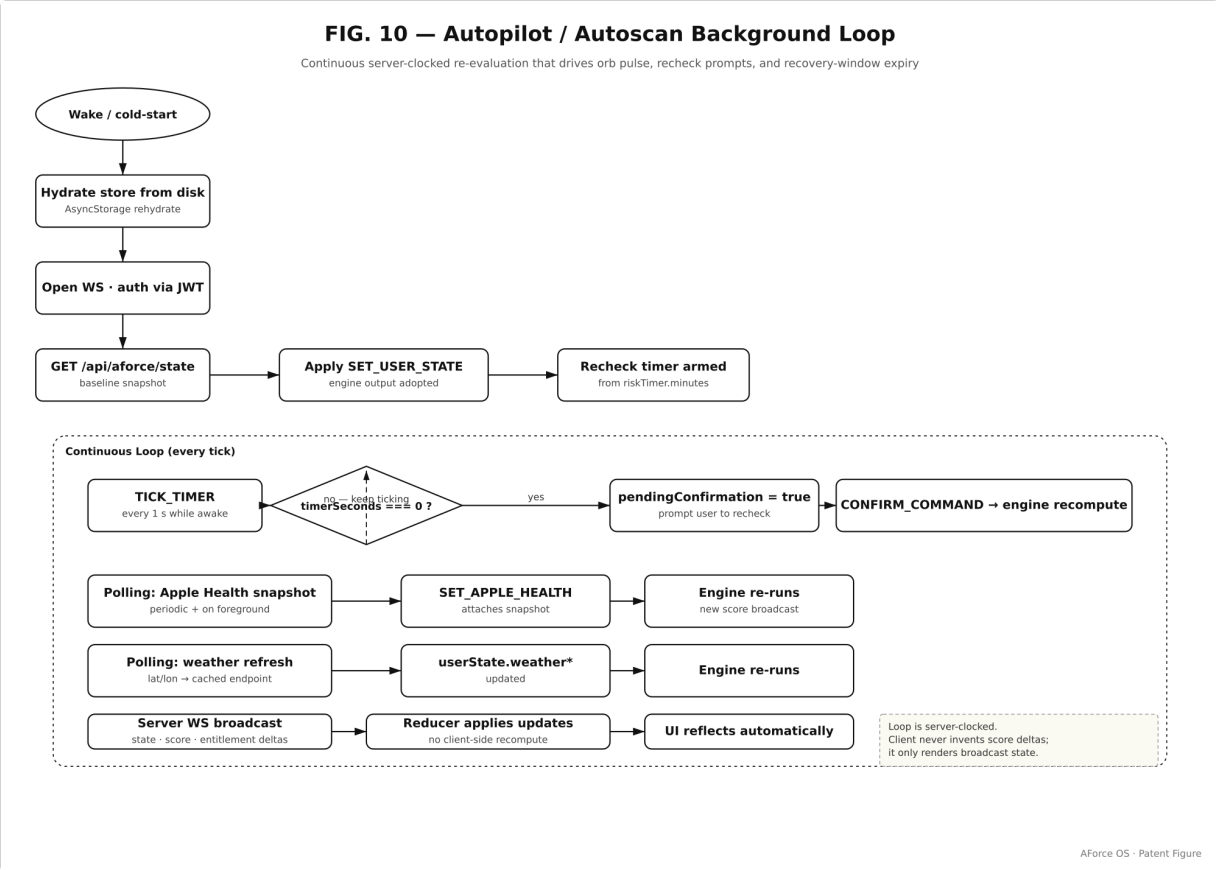


FIG. 10 — Autopilot / Autoscan. Top half: sweat session → deficit % → autopilot cadence ladder ($\geq 4\%$ → 8 min / critical, $\geq 2\%$ → 12 min / high, else → 20 min / moderate, 4 h window) → atomic state transition that simultaneously stores the autopilot tuple, resets the visible recheck timer, and clears any pending confirmation prompt. Bottom half: barcode / QR scan → product recognition (local catalog → Open Food Facts fallback) → live-state fit score → conditional AForce replacement decision when brand fit exceeds scanned fit by > 4 points.

Cross-reference

Figure	Brief section	Implementation file(s)
FIG. 1	§1	artifacts/aforce-os/ , artifacts/api-server/
FIG. 2	§1.1	app/(tabs)/index.tsx , components/StatusPulseOrb.tsx
FIG. 3	§1.4	hooks/useEntitlement.ts , app/subscription.tsx
FIG. 4	§3.3	utils/scoringEngine.ts:458-531 , services/bacEstimationService.ts
FIG. 5	§1.4	services/voiceService.ts , services/voiceTemplateEngine.ts
FIG. 6	§1.5	live canvas iframes + screens/*
FIG. 7	§1.2	app/_layout.tsx , store/useAppStore.tsx
FIG. 8	§2	utils/scoringEngine.ts , services/hydrationScoreService.ts
FIG. 9	§3	utils/scoringEngine.ts:537-591 , :728-748 , :790-838
FIG. 10	§4, §5	services/sweatRateEngine.ts:489-493 , services/hydrationScanService.ts

End of figures. Source vectors: attached_assets/patent/fig{1..10}-*.svg .